

Notes on Frobenius Algebras, Dialogue Categories and Tensorial Ribbon Logic

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Contents

1	Frobenius-related concepts	1
2	Dialogue Categories and Chiralities	14
3	Categorical Bimodules	16
4	Frobenius Pseudomonoids	16
5	Frobenius Amphimonoids	20
6	Ribbon Tensorial Logic: basic remarks	25
7	On Balanced Dialogue Categories and Ribbon Tensorial Logic	31
8	The Sequent Calculus and its commuting conversions	33
9	Focusing Theorem	36
10	Proof-as-Tangle theorem	38

1 Frobenius-related concepts

Let n be a non-zero natural number. The category $\mathbf{Cob}(n)$ is composed of the following data:

- The objects of $\mathbf{Cob}(n)$ are closed orientated $n - 1$ -dimensional manifolds.
- If $M, N \in \mathbf{Cob}(n)$ are *bordisms* from M to N . In turn, a bordism from M to N is an n -dimensional manifold B such that there is a diffeomorphism:

$$\partial B \cong (-M) \cup N$$

where ∂B is the boundary of B , $-M$ is the manifold M with the reversed orientation. $B = B'$ are identified whenever the following holds:

$$\partial B \cong (-M) \cup N \cong \partial B'$$

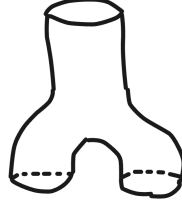
- If $M \in \mathbf{Cob}(n)$, then the morphism id_M is given by the trivial bordism $B = M \otimes [0, 1]$.
- The composition of $B : M \rightarrow N$ and $B' : N \rightarrow P$ is given by the glueing the corresponding bordisms.

$\mathbf{Cob}(n)$ is endowed with the structure of a symmetric monoidal category, where $M \otimes N$ is defined by taking the disjoint union $M \sqcup N$ and the empty manifold as the identity I .

The idea of topological field theory is to construct a symmetric monoidal functor $\mathbf{Cob}(n) \rightarrow \mathbf{Vect}_k$. The question is what information is contained in a topological field theory of dimension n . In the case of $n = 2$, a topological field theory is the same thing as a commutative and cocommutative Frobenius algebra in $\mathbf{Vect}_k$. A Frobenius algebra is a Frobenius monoid in $\mathbf{Vect}_k$.

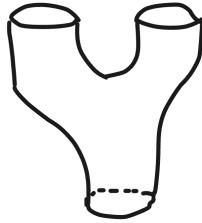
Definition 1.1. Let $\mathcal{V}$ be a monoidal category, a bimonoid A in $\mathcal{V}$ equipped with the structures of monoid and comonoid (A, m, e) and (A, d, u) respectively. That is, A is equipped with a binary operation $m : A \otimes A \rightarrow A$:

$$A \otimes A \xrightarrow{m} A$$

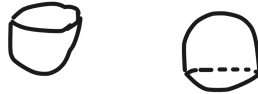


and a binary co-operation $d : A \rightarrow A \otimes A$:

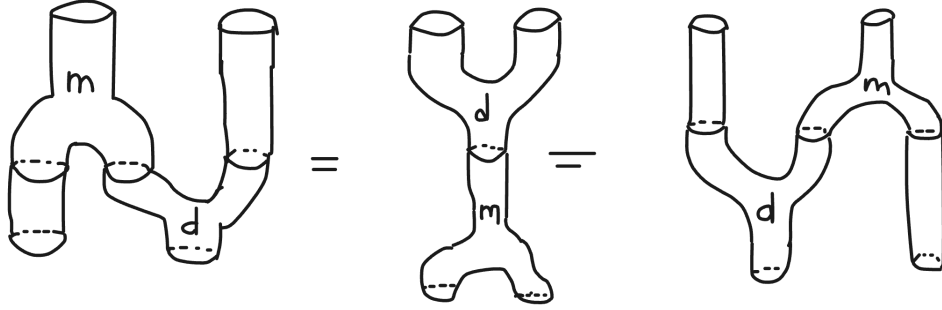
$$A \xrightarrow{d} A \otimes A$$



such that they are associative and co-associative respectively, and A is also equipped with a unit $e : I \rightarrow A$ and a counit $A \rightarrow I$:



A bimonoid is *Frobenius* if the following equalities are satisfied:



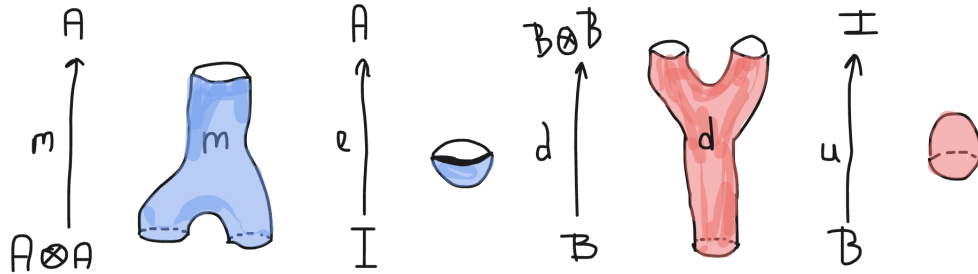
A Frobenius monoid is commutative (cocommutative) if its monoid (comonoid) component is commutative (cocommutative).

The characterisation of topological field theories of dimension 2 is generalisable to an arbitrary symmetric monoidal category.

Proposition 1.1. Let $\mathcal{V}$ be a symmetric monoidal category. There is a 1-to-1 correspondence symmetric monoidal functor $\mathbf{Cob}(2) \rightarrow \mathcal{V}$ and commutative and cocommutative Frobenius monoids.

Proof. See [Koc04]. □

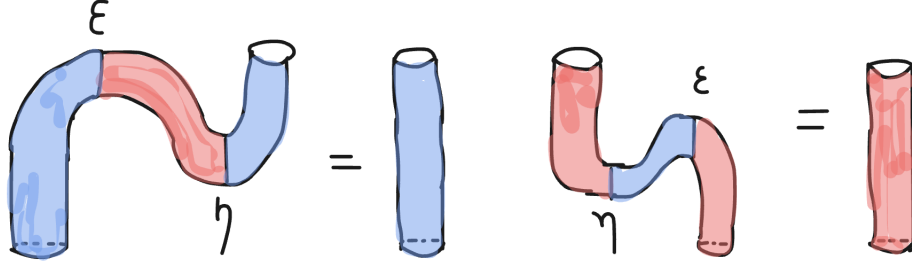
The further idea is to study Frobenius monoids in a more logical framework inspired by game semantics. We decouple the monoid side (A, m, e) from the comonoid side (B, d, u) in the definition of a Frobenius monoid. Different carriers describe the “split personality” aspect of the Frobenius monoid. We differ operations from cooperations by the blue and red colours:



The question is how A and B are coupled inside a Frobenius monoid. First of all, A and B should be involved in *exact pairing* $A \dashv B$ defined as a pair of morphisms:

- $\eta : I \rightarrow B \otimes A$
- $\varepsilon : A \otimes B \rightarrow I$

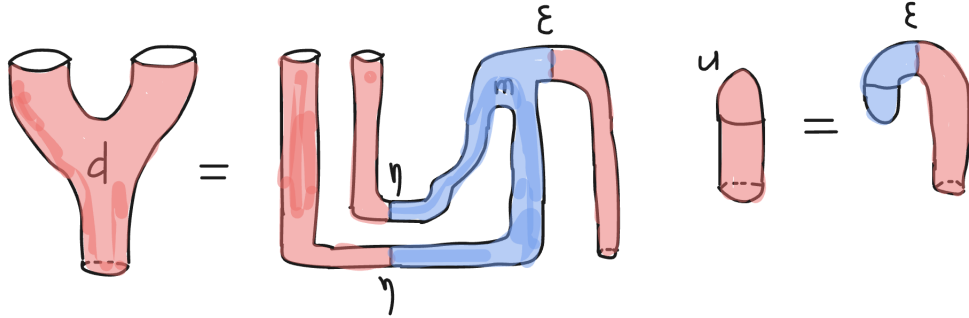
satisfying the following zig-zag equations:



If $\mathcal{V}$ is the category of k -vector spaces, one can equip A and B with a non-degenerate form $\varepsilon : A \otimes B \rightarrow k$. The exact pairing $A \dashv B$ should be compatible with the comonoid and monoid structures the following way. We define a monoid-comonoid pairing:

$$(A, m, e) \dashv (B, d, u)$$

between a monoid and a comonoid as an exact pairing between carriers $A \dashv B$ satisfying the following two equalities:



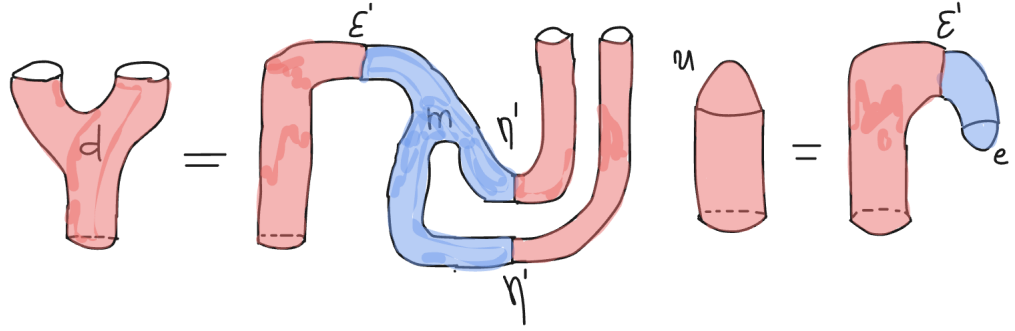
Those equations mean that the comonoid structure (B, d, u) can be recovered from the monoid structure (A, m, e) on A and the other way round. Symmetrically, the existence of a monoid-comonoid pairing

$$(B, d, u) \dashv (A, m, e)$$

defined as an exact pairing $B \dashv A$ between the underlying objects with the components

- $\eta' : I \rightarrow B \otimes A$
- $\varepsilon' : A \otimes B \rightarrow I$

satisfying the following equations:



This leads to the definition of a Frobenius pair:

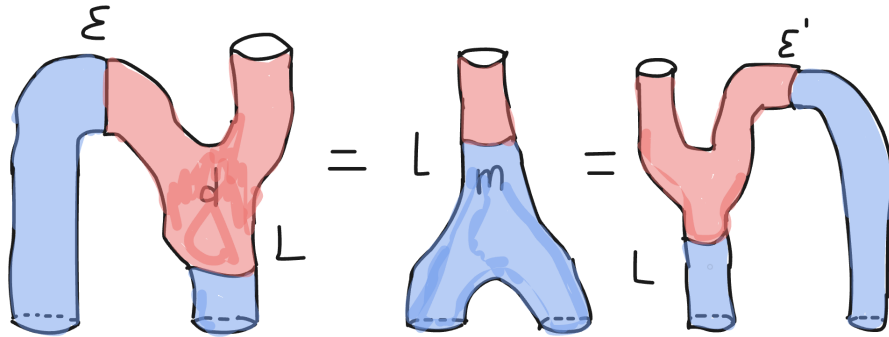
Definition 1.2. A Frobenius pair in a symmetric monoidal category $\mathcal{V}$ consists of a monoid-comonoid exact pairing:

$$(A, m, e) \dashv (B, d, u)$$

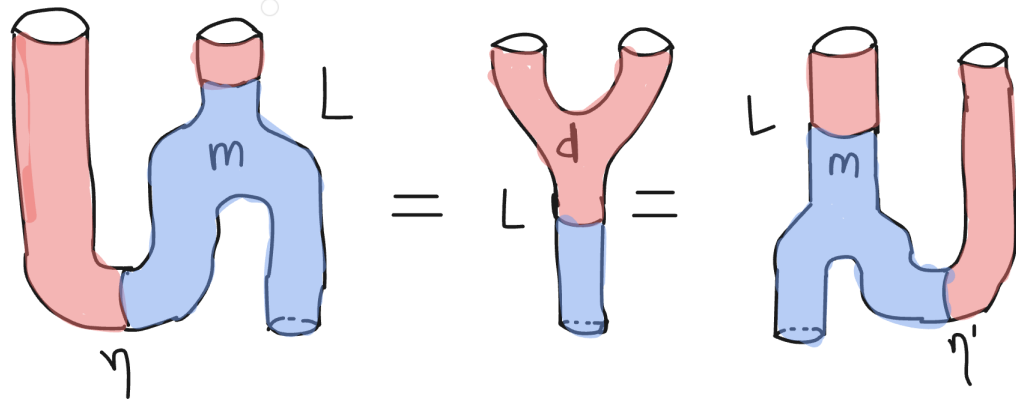
and a comonoid-monoid pairing

$$(B, d, u) \dashv (A, m, e)$$

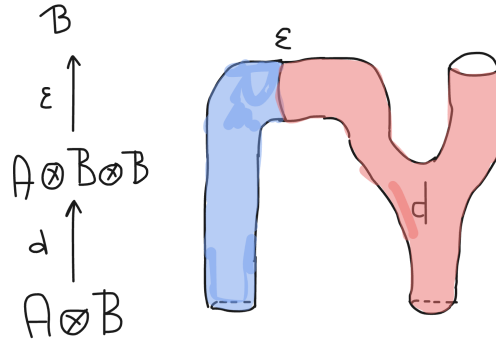
along with an isomorphism $L : A \cong B$ such that the following is satisfied:



The above equations have the following reformulation:



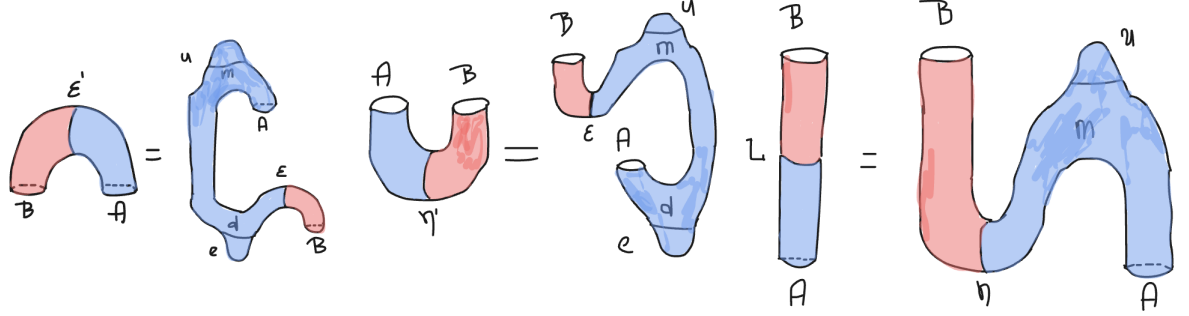
These equalities have the following interpretation. The monoid-comonoid pairing $(A, m, e) \dashv (B, d, u)$ induces a left action of the monoid (A, m, e) on B :



The equation from the definition of a Frobenius pair means that the morphism L transports the left action of the monoid A on itself into the action of A on the comonoid carrier B . Symmetrically, the exact pairing $B \dashv A$ a right action of the monoid A on B , so the second equation from the definition of a Frobenius pair asks that L transports the canonical right action of A on itself into the right action of A on B . There is the following correspondence between Frobenius pairs and Frobenius monoids.

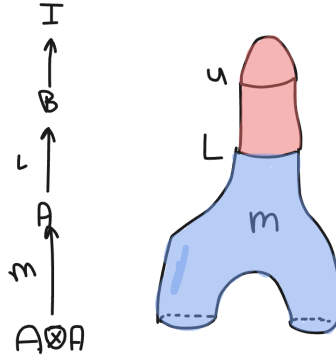
Proposition 1.2. Let $\mathcal{V}$ be a monoidal category, then there is a 1-to-1 correspondence between Frobenius pairs (A, B) and Frobenius monoids A equipped with an exact pairing $A \dashv B$.

Proof. Given a Frobenius monoid A with an exact pairing $A \dashv B$ and unit η and counit ε , then one defines the unit η' and the counit ε' of the exact pairing $B \dashv A$ and the isomorphism L the following way:



B inherits the comonoid structure (B, d, u) from the monoid structure (A, m, e) and the exact pairing $A \dashv B$. Then one checks that the comonoid structure coincides with the comonoid structure induced by the exact pairing $B \dashv A$. Finally, one checks the equalities from the definition of a Frobenius pair. Conversely, any Frobenius pair (A, B) induces a Frobenius monoid on A with the monoid structure (A, m, e) and the comonoid structure (A, d', u') induced with the isomorphism with (B, d, u) by L . $\square$

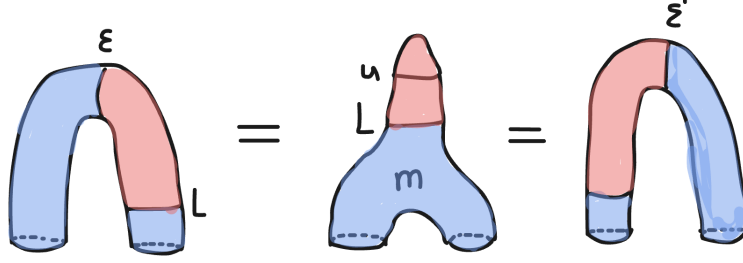
Let (A, B) be a Frobenius pair in a monoidal category $\mathcal{V}$, the following morphism



defined a bilinear form $||\cdot, \cdot|| : A \otimes A \rightarrow I$ called the *Frobenius bracket*. The definition of a Frobenius bracket ensures that the following condition is satisfied:

$$||a_1 \bullet a_2, a_3|| = ||a_1, a_2 \bullet a_3||$$

Using the definition of a Frobenius pair, one can reformulate the definition of a Frobenius brackets the following way:



A *braiding* for a monoidal category $\mathcal{V}$ is a family of natural isomorphisms:

$$\gamma_{A,B} : A \otimes B \rightarrow B \otimes A$$

such that the following diagrams commute:

$$\begin{array}{ccccc}
 & A \otimes (B \otimes C) & \xrightarrow{\gamma_{A,B \otimes C}} & (B \otimes C) \otimes A & \\
 a_{A,B,C} \nearrow & & & & \searrow a_{B,C,A} \\
 (A \otimes B) \otimes C & & & & B \otimes (C \otimes A) \\
 \gamma_{A,B} \otimes 1_C \searrow & & & & \nearrow \gamma_{A,C} \otimes 1 \\
 (B \otimes A) \otimes C & \xrightarrow{a_{B,A,C}} & B \otimes (A \otimes C) & & \\
 \\
 & (A \otimes B) \otimes C & \xrightarrow{\gamma_{A \otimes B,C}} & C \otimes (A \otimes B) & \\
 a_{A,B,C}^{-1} \nearrow & & & & \searrow a^{-1}_{C,A,B} \\
 A \otimes (B \otimes C) & & & & (C \otimes A) \otimes B \\
 1 \otimes \gamma_{B,C} \searrow & & & & \nearrow \gamma_{A,C} \otimes 1 \\
 A \otimes (C \otimes B) & \xrightarrow{a_{A,C,B}^{-1}} & (A \otimes C) \otimes B & &
 \end{array}$$

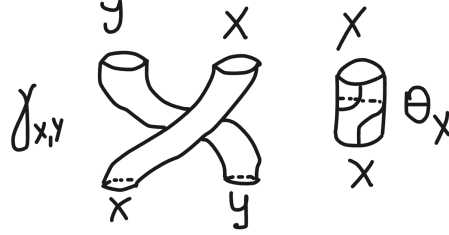
A *braided monoidal category* is a monoidal category with a choice of braiding.

Let $\mathcal{V}$ be a braided monoidal category, a *twist* for $\mathcal{V}$ is a family of isomorphisms $\theta_A : A \rightarrow A$ such that $\theta_I = 1_I$ and

$$\begin{array}{ccc}
 A \otimes B & \xrightarrow{\gamma_{A,B}} & B \otimes A \\
 \theta_{A \otimes B} \downarrow & & \downarrow \theta_B \otimes \theta_A \\
 A \otimes B & \xleftarrow{\gamma_{B,A}} & B \otimes A
 \end{array}$$

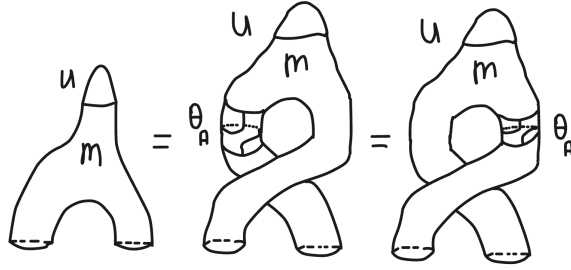
A *balanced monoidal category* is a braided monoidal category with a choice of a twist.

Typically, a balanced monoidal category $\mathcal{V}$ is given as $\mathcal{V} = \mathbf{Mod}(H)$, the category of representations of a quantum group H . The braiding γ and the twist θ are represented as follows:

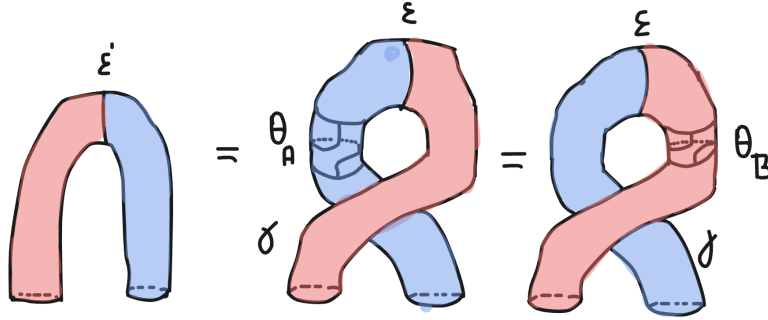


The twist θ_X is understood as the operation of applying a rotation of angle 2π on the border X of the two-dimensional manifold. The extra structure allows formulating the definition of helical Frobenius monoid.

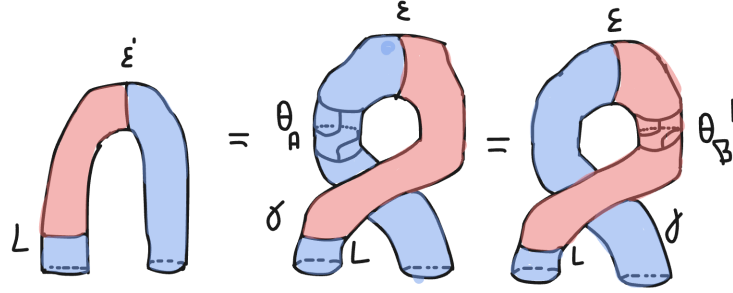
Definition 1.3. A Frobenius monoid A in a balanced monoidal category $\mathcal{V}$ is called *helical* if the following is satisfied:



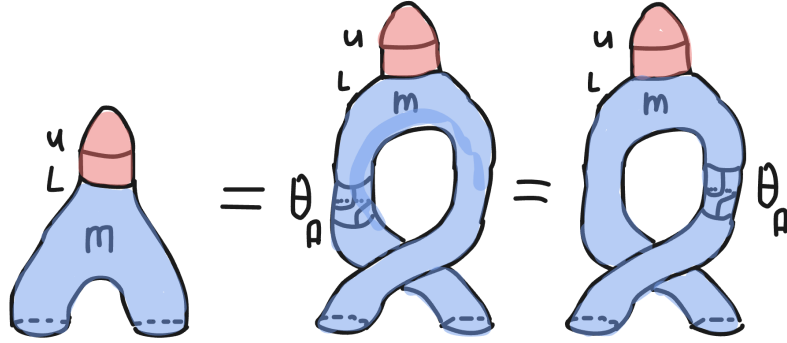
Definition 1.4. A Frobenius pair (A, B) in a balanced monoidal category is *helical* if the following equalities are satisfied:



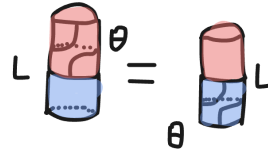
One can reword the definition of helicity of a Frobenius pair the following way using the reversibility of L :



In turn, the above condition is equivalent to the condition claiming that the Frobenius bracket is commutative in the sense of the following equalities:



The above equivalence, in turn, follows from the fact that the twist operation is a natural isomorphism of the identity functor with itself, and, therefore, the following is satisfied:



So we rephrase the condition the following algebraic way:

$$||a_1, a_2|| = ||a_2, a_1||$$

Thus we have the following cycling property of the Frobenius bracket:

$$||a_1 \bullet a_2, a_3|| = ||a_3 \bullet a_1, a_2|| = ||a_2 \bullet a_3, a_1||.$$

Proposition 1.3. Let $\mathcal{V}$ be a balanced monoidal category, then there is a 1-to-1 correspondence between helical Frobenius pairs and helical Frobenius monoids A equipped with exact pairing $A \dashv B$.

Definition 1.5. Let $\mathcal{V}$ be a balanced monoidal category, then $\mathcal{V}$ is *ribbon* if each $A \in \mathcal{V}$ is equipped with an exact pairing $A \dashv A^*$ whose counit $\varepsilon_A : A \otimes A^* \rightarrow I$ satisfies the following equation:

The diagram shows an equality between two string diagrams. On the left, a cup-shaped string labeled A is connected to a cap-shaped string labeled A^* . The cup has a label θ_A on its left side and the cap has a label ε_A on its right side. On the right, the same configuration is shown, but the cup is labeled A and the cap is labeled A^* , with the label ε_A on the left side of the cup and θ_{A^*} on the right side of the cap.

From this definition, we have that if the right dual A^* of A is also a left dual of A , when counit ε'_A of the exact pairing $A^* \dashv A$ defined the following way:

The diagram shows an equality between two string diagrams. On the left, a cup-shaped string labeled A^* is connected to a cap-shaped string labeled A . The cup has a label ε'_A on its top side. On the right, the same configuration is shown, but the cup is labeled A^* and the cap is labeled A , with the label ε_A on the top side of the cup and θ_{A^*} on the right side of the cap. A crossing symbol γ is shown between the two diagrams.

Also, any Frobenius pair in a ribbon category $\mathcal{V}$ satisfies the following:

The diagram shows an equality between two string diagrams. On the left, a cup-shaped string labeled A is connected to a cap-shaped string labeled B . The cup has a label θ_A on its left side and the cap has a label ε on its right side. On the right, the same configuration is shown, but the cup is labeled A and the cap is labeled B , with the label ε on the left side of the cup and θ_B on the right side of the cap.

One can also observe the opposite category $\mathcal{V}^{op(0,1)}$ of a balanced monoidal category $\mathcal{V}$ is also balanced, the twist and the braiding operations are inherited from $\mathcal{V}$. By $\mathcal{V}^{op(0,1)}$, we mean that the orientations of the tensor product and morphism are reserved. The transformation $\mathcal{V} \mapsto \mathcal{V}^{op(0,1)}$ consists in applying a central symmetry to string diagrams. The exact pairing $A \dashv A^*$ induces a monoidal functor $(.) : \mathcal{V} \rightarrow \mathcal{V}^{op(0,1)}$ transporting the ribbon structure to the ribbon structure. Therefore, every exact pairing $A \dashv A^*$ in $\mathcal{V}$ defines an exact pairing $A \dashv A^*$ in $\mathcal{V}^{op(0,1)}$ with unit defined as the image $\varepsilon_A^* : I \rightarrow A^* \otimes^{op} A$ of the counit ε_A . Therefore, a ribbon structure on $\mathcal{V}$ induces the ribbon structure on $\mathcal{V}^{op(0,1)}$.

Let (A, B) a Frobenius pair in a ribbon category. The monoidal structure of $(.)^*$ sends the comonoid structure on B to the monoid structure on B^* .

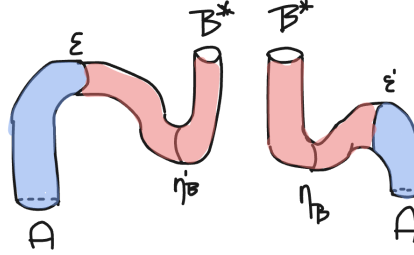
The resulting monoid structure (B^*, d^*, u^*) can be constructed from either the exact pairing $B^* \dashv B$ or from the exact pairing $B \dashv B^*$. Therefore, the monoid (B^*, d^*, u^*) is involved in two exact pairing with the comonoid (B, d, u) as a left and as a right dual:

$$\begin{aligned} (B^*, d^*, u^*) \dashv (B, d, u) \\ (B, d, u) \dashv (B^*, d^*, u^*) \end{aligned}$$

with unit η_B and counit ε_B and unit η'_B and counit ε'_B respectively. Let us compare those exact pairings between (A, m, e) and (B, d, u) involved in the definition of a Frobenius pair:

$$\begin{aligned} (A, m, e) \dashv (B, d, u) \\ (B, d, u) \dashv (A, m, e) \end{aligned}$$

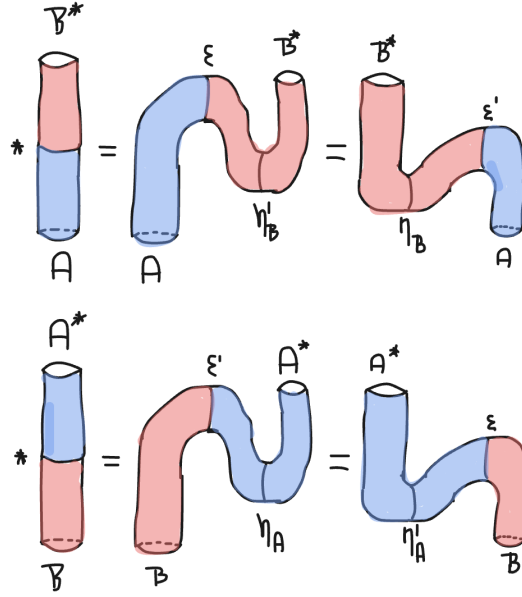
with unit η and counit ε and unit η' and counit ε' respectively. Each comparison induces a monoid isomorphism of (A, m, e) with (B^*, d^*, u^*) defined as:



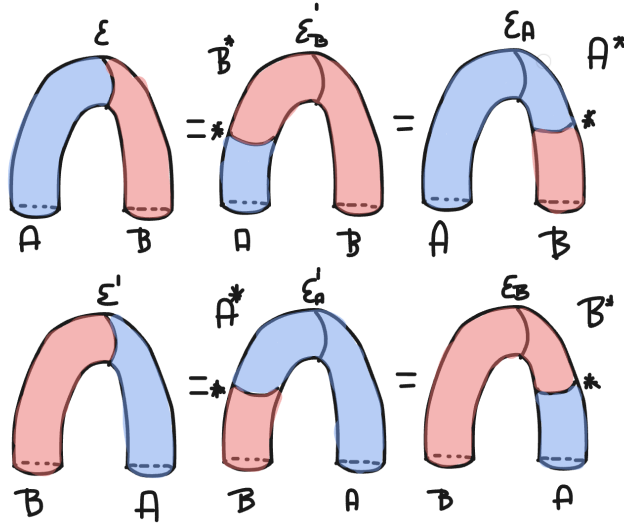
The point is that the two isomorphisms $A \rightarrow B^*$ coincide exactly when a Frobenius pair (A, B) is helical. In this case, one obtains the following isomorphisms:

$$(A, m, e) \begin{array}{c} \xrightarrow{(\cdot)^*} \\ \xleftarrow{*(\cdot)} \end{array} (B^*, d^*, u^*)$$

where $(\cdot)^* : A \rightarrow B^*$ and $*(\cdot) : B^* \rightarrow A$ are isomorphisms defined as:

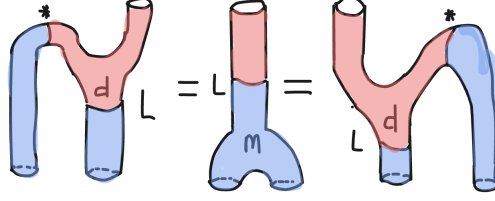


From the definitions, we also conclude the following equalities:



Those equalities imply the following equivalent definition

Proposition 1.4. A helical Frobenius pair (A, B) in a ribbon category is the same thing as a monoid (A, m, e) and a comonoid (B, d, u) equipped with a monoid isomorphism $(.)^* : (A, m, e) \rightarrow (B^*, d^*, u^*)$ and an isomorphism $L : A \rightarrow B^*$ such that the following equalities are satisfied:



2 Dialogue Categories and Chiralities

Definition 2.1. A dialogue category is a monoidal category $(\mathcal{C}, \otimes, e)$ equipped with an object $\perp$ along with a family of isomorphisms

$$\varphi_{x,y} : \mathcal{C}(x \otimes y, \perp) \cong \mathcal{C}(y, x \multimap \perp)$$

natural in x and y ; along with a family of isomorphisms

$$\psi_{x,y} : \mathcal{C}(x \otimes y, \perp) \cong \mathcal{C}(x, \perp \multimap y)$$

natural in x and y as well.

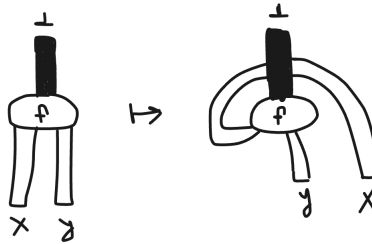
Definition 2.2. A helical dialogue category is a dialogue category $\mathcal{C}$ equipped with a family of bijections natural in x and y

$$\text{wheel}_{x,y} : \mathcal{C}(x \otimes y, \perp)$$

such that the following diagram commutes for each $x, y, z \in \mathcal{C}$:

$$\begin{array}{ccc}
 \mathcal{C}((y \otimes z) \otimes x, \perp) & \xrightarrow{\cong} & \mathcal{C}(y \otimes (z \otimes x), \perp) \\
 \uparrow \text{wheel}_{x,y \otimes z} & & \downarrow \text{wheel}_{y,z \otimes x} \\
 \mathcal{C}(x \otimes (y \otimes z), \perp) & & \mathcal{C}((z \otimes x) \otimes y, \perp) \\
 \downarrow \cong & & \uparrow \cong \\
 \mathcal{C}((x \otimes y) \otimes z, \perp) & \xrightarrow{\text{wheel}_{x \otimes y, z}} & \mathcal{C}(z \otimes (x \otimes y), \perp)
 \end{array}$$

Graphically, $\text{wheel}_{x,y}$ is visualised as:



Definition 2.3. A helical chirality is a pair of monoidal categories $(\mathcal{A}, \otimes, true)$ and $(\mathcal{B}, \otimes, false)$ equipped with a monoidal equivalence and an adjunction respectively:

$$\begin{array}{ccc} \mathcal{A} & \xrightleftharpoons[\ast(\cdot)]{(\cdot)^\ast} & \mathcal{B} \\ \mathcal{A} & \xrightleftharpoons[R]{L} & \mathcal{B} \end{array}$$

with the following isomorphisms:

$$\begin{aligned} \chi_{m,a,b}^L &: \langle m \otimes a \mid b \rangle \rightarrow \langle a \mid m^\ast \otimes b \rangle \\ \chi_{m,a,b}^R &: \langle a \otimes m \mid b \rangle \rightarrow \langle a \mid b \otimes m^\ast \rangle \end{aligned}$$

where $\langle \cdot \mid \cdot \rangle := \mathcal{A}(\cdot, R(\cdot)) : \mathcal{A}^{op} \times \mathcal{B} \rightarrow Set$ is the evaluation bracket. The curried combinators χ^L and χ^R satisfy the following diagrams for each $m, n, a \in \mathcal{A}$ and for $b \in \mathcal{B}$:

$$\begin{array}{ccccc} \langle (m \otimes n) \otimes a \mid b \rangle & \xrightarrow{\chi_{m \otimes n}^L} & & & \langle a \mid (m \otimes n)^\ast \otimes b \rangle \\ \cong \downarrow & & & & \uparrow \cong \\ \langle m \otimes (n \otimes a) \mid b \rangle & \xrightarrow{\chi_m^L} \langle n \otimes a \mid m^\ast \otimes b \rangle & \xrightarrow{\chi_n^L} & & \langle a \mid n^\ast \otimes m^\ast \otimes b \rangle \\ \\ \langle a \otimes (m \otimes n) \mid b \rangle & \xrightarrow{\chi_{m \otimes n}^R} & & & \langle a \mid b \otimes (m \otimes n)^\ast \rangle \\ \cong \downarrow & & & & \uparrow \cong \\ \langle (a \otimes m) \otimes n \mid b \rangle & \xrightarrow{\chi_n^R} \langle a \otimes m \mid b \otimes n^\ast \rangle & \xrightarrow{\chi_m^R} & & \langle a \mid (b \otimes n^\ast) \otimes m^\ast \rangle \\ \\ \langle (m \otimes a) \otimes n \mid b \rangle & \xrightarrow{\chi_n^R} \langle m \otimes a \mid b \otimes n^\ast \rangle & \xrightarrow{\chi_m^R} & & \langle a \mid m^\ast \otimes (b \otimes n^\ast) \rangle \\ \cong \downarrow & & & & \downarrow \cong \\ \langle m \otimes (a \otimes n) \mid b \rangle & \xrightarrow{\chi_m^L} \langle a \otimes n \mid m^\ast \otimes b \rangle & \xrightarrow{\chi_n^R} & & \langle a \mid (m^\ast \otimes b) \otimes n^\ast \rangle \end{array}$$

Every helical dialogue category $\mathcal{C}$ induces defines a helical dialogue chirality by taking $\mathcal{A} = \mathcal{C}$, $\mathcal{B} = \mathcal{C}^{op(0,1)}$, $La := a \multimap \perp$, $Rb := \perp \multimap b$. The right curriedification of the combinator χ^R is simply defined by using the dialogue structure of the category $\mathcal{C}$:

$$\langle a \otimes m \mid b \rangle = \mathcal{C}(a \otimes m, \perp \multimap b) \xrightarrow{\psi_{a \otimes m, b}^{-1}} \mathcal{C}(a \otimes m \otimes b, \perp) \xrightarrow{\psi_{a, m \otimes b}} \mathcal{C}(a, \perp \multimap m \otimes b) = \langle a \mid b \otimes m^\ast \rangle$$

The left curriedification χ^L requires the helical structure:

$$\begin{aligned} \langle m \otimes a \mid b \rangle &= \mathcal{C}(m \otimes a, \perp \multimap b) \xrightarrow{\psi_{m \otimes a, b}^{-1}} \mathcal{C}(m \otimes a \otimes b, \perp) \xrightarrow{\text{wheel}_{m, a \otimes b}} \\ &\mathcal{C}(a \otimes b \otimes m, \perp) \xrightarrow{\psi_{m \otimes b, a}} \mathcal{C}(a, \perp \multimap b \otimes m) = \langle a \mid m^\ast \otimes b \rangle \end{aligned}$$

3 Categorical Bimodules

Let $\mathcal{A}$ and $\mathcal{B}$ be categories, an $\mathcal{AB}$ -module M is defined as a bifunctor $M : \mathcal{A}^{op} \times \mathcal{B} \rightarrow \mathbf{Set}$. The bicategory (a weak 2-category) **BiMod** of bimodules consists of the following data:

- small categories as 0-cells,
- $\mathcal{AB}$ -modules M as 1-cells $M : \mathcal{A} \rightarrow \mathcal{B}$,
- natural transformations $\theta : N \Rightarrow M : \mathcal{A}^{op} \times \mathcal{B} \rightarrow \mathbf{Set}$ as 2-cells $\theta : M \Rightarrow N : \mathcal{A} \rightarrow \mathcal{B}$.

The orientation of 2-cells allows defining a 2-monoidal functor $\mathbf{Cat} \rightarrow \mathbf{BiMod}$ transporting every functor $F : \mathcal{A} \rightarrow \mathcal{B}$ to the bimodule:

$$F_{\bullet} : (a, b) \mapsto \mathcal{A}(Fa, b)$$

One can view **BiMod** as a 2-dimensional Kleisli construction on the small limit completion $\mathcal{C} \mapsto [\mathcal{C}, \mathbf{Set}]^{op}$. One can recover the more familiar convention corresponding to the small colimit completion $\mathcal{C} \mapsto [\mathcal{C}^{op}, \mathbf{Set}]$ by taking the bicategory $\mathbf{BiMod}^{op(1,2)}$ obtained from **BiMod** by reversing 1 and 2-cells.

There is also a monoidal 2-functor $\mathbf{Cat} \rightarrow \mathbf{BiMod}^{op(1,2)}$ defined as follows. First of all, the operation $(.)^{op} : \mathcal{C} \mapsto \mathcal{C}^{op}$ defines a 2-functor $\mathbf{Cat} \rightarrow \mathbf{Cat}^{op(1,2)}$. **BiMod** is also symmetric monoidal with $\otimes = \times$. It is also autonomous, and its ribbons are trivial. There exists a 2-monoidal functor $(.)^* : \mathbf{BiMod} \rightarrow \mathbf{BiMod}^{op(0,1)}$ transporting every A to its dual A^* . A^* happens to coincide with A^{op} . Therefore, one has a 2-monoidal functor:

$$\mathbf{Cat} \xrightarrow{op} \mathbf{Cat}^{op(0,1)} \rightarrow \mathbf{BiMod}^{op(2)} \xrightarrow{*} \mathbf{BiMod}^{op(1,2)}$$

transporting every category $\mathcal{A}$ to itself and every $F : \mathcal{A} \rightarrow \mathcal{B}$ to the bimodule

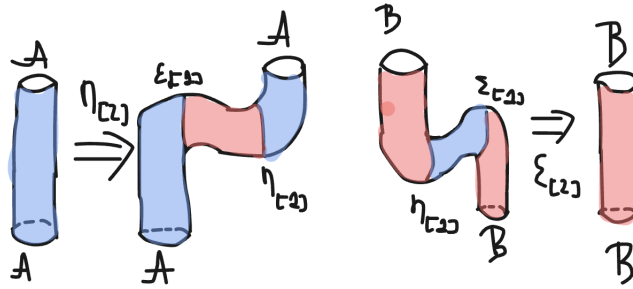
$$F^{\bullet} : (a, b) \mapsto \mathcal{A}(a, Fb)$$

$F^{\bullet}$ is also left adjoint to $F_{\bullet}$ in **BiMod**.

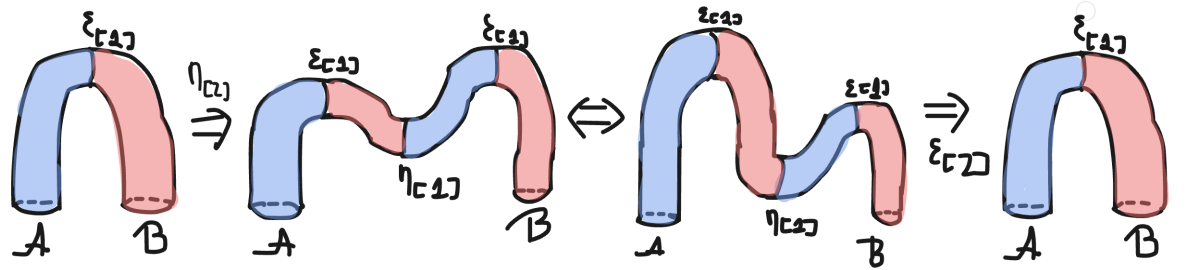
4 Frobenius Pseudomonoids

The concept of a Frobenius pseudomonoid reflects the properties of a dialogue category $\mathcal{A}$ transported from **Cat** to the monoidal bicategory **BiMod**.

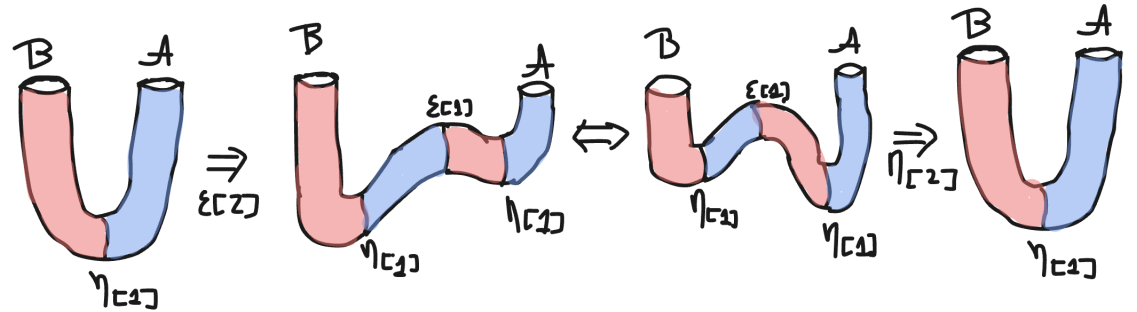
Definition 4.1. A lax pairing $\mathcal{A} \dashv \mathcal{B}$ in a monoidal bicategory consists of a pair of 1-cells $\eta_{[1]} : \mathcal{A} \otimes \mathcal{B} \rightarrow I$ and $\varepsilon_{[1]} : \mathcal{B} \otimes \mathcal{A} \rightarrow I$ along with a pair of 2-cells:



such that the following composite 2-cell

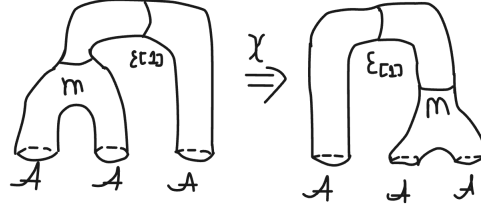


coincides with the identity on the 1-cell $\varepsilon_{[1]}$ and, symmetrically, the opposite 2-cell:

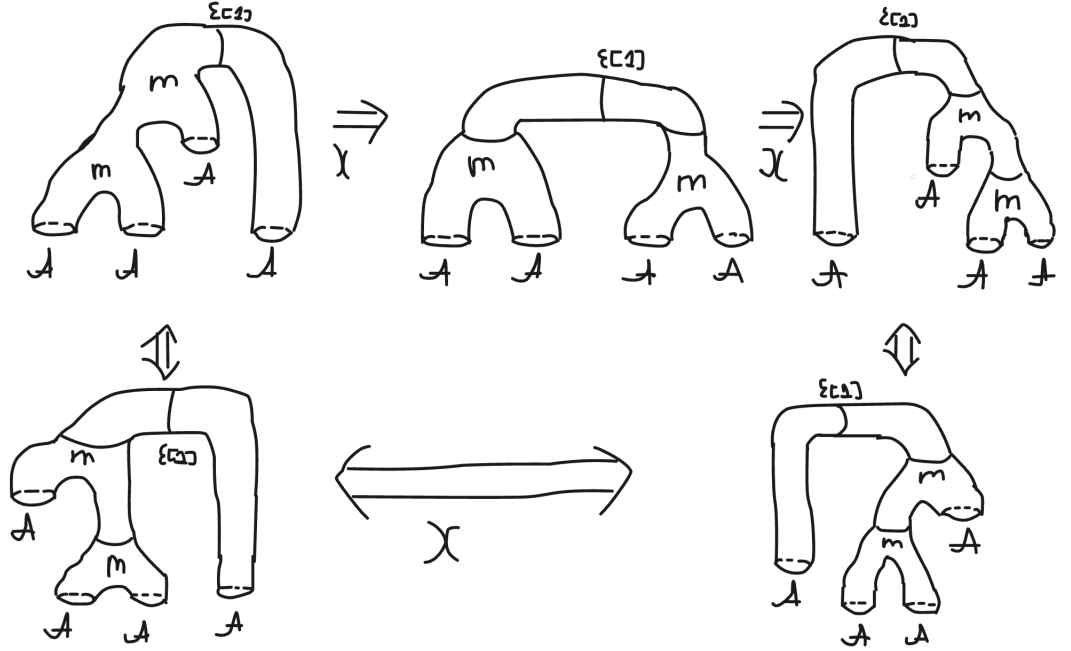


coincides with the identity on the 1-cell $\eta_{[1]}$.

Definition 4.2. A *Frobenius form* on a pseudomonoid $\mathcal{A}$ is a monoidal bicategory is a lax pairing $\mathcal{A} \dashv \mathcal{A}$ equipped with a 2-cell



called *the associativity law of a Frobenius form* such that the following variation of the MacLane pentagon is satisfied



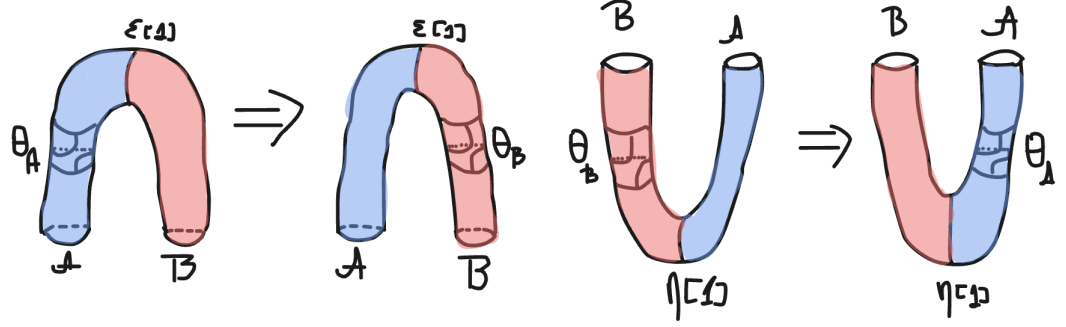
A *Frobenius pseudomonoid* is a pseudomonoid equipped with a Frobenius form.

Observe that in **BiMod**, every dialogue category $\mathcal{A}$ defined a Frobenius pseudomonoid with Frobenius form defined as:

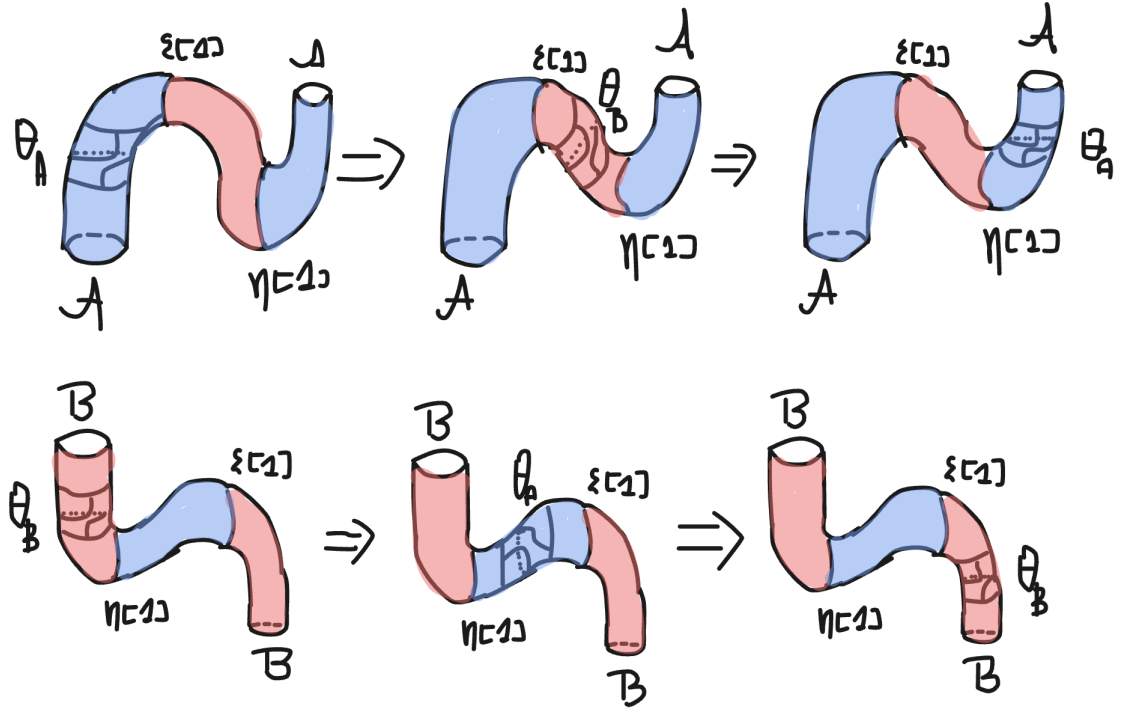
$$\begin{aligned}\varepsilon_{[1]} &: (a_1, a_2) \mapsto \mathcal{A}(a_1 \otimes a_2, \perp) \\ \eta_{[1]} &: (a_1, a_2) \mapsto \mathcal{A}(\perp \multimap a_1, a_2)\end{aligned}$$

and χ is defined by the associativity law from $\mathcal{A}$. But the above notion of Frobenius pseudomonoid does not coincide with the notion of a Frobenius monoid when the underlying monoidal bicategory is a monoidal category with trivialised 2-cells.

Definition 4.3. A ribbon structure on a lax pairing $\mathcal{A} \dashv \mathcal{B}$ in a balanced monoidal bicategory is a pair of invertible 2-cells



such that both composite 2-cells

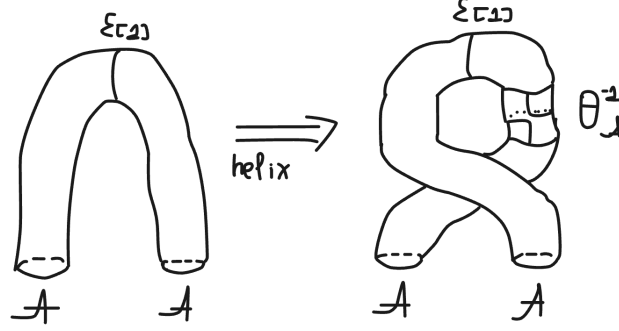


coincide with the identity 2-cells.

A *lax ribbon pairing* is a lax pairing equipped with a pairing structure.

Definition 4.4. A helical Frobenius pseudomonoid in a balanced monoidal

category is a Frobenius pseudomonoid whose lax pairing is equipped with a ribbon structure and the following invertible 2-cell:



helix can be viewed as a 2-cell $\|a_1, a_2\| \Rightarrow \|a_2, a_1\|$ such that the following commutes:

$$\begin{array}{ccccc}
 \|a_1 \bullet a_2, a_3\| & \xrightarrow{\chi} & \|a_1, a_2 \bullet a_3\| & \xrightarrow{\text{helix}} & \|a_2 \bullet a_3, a_1\| \\
 \Downarrow \text{helix} & & & & \Downarrow \chi \\
 \|a_3, a_1 \bullet a_2\| & \xleftarrow{\chi} & \|a_3 \bullet a_1, a_2\| & \xleftarrow{\text{helix}} & \|a_2, a_3 \bullet a_1\|
 \end{array}$$

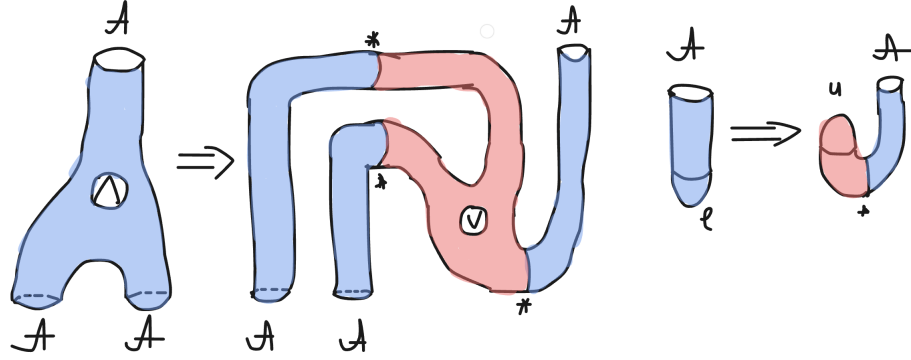
Every helical dialogue category $\mathcal{A}$ induces a Frobenius pseudomonoid in **BiMod** with helix defined as follows:

$$\text{helix}_{a_1, a_2} := \text{wheel}_{a_1, a_2}^{-1} : \mathcal{A}(a_2 \otimes a_1, \perp) \Rightarrow \mathcal{A}(a_1 \otimes a_2, \perp)$$

5 Frobenius Amphimonoids

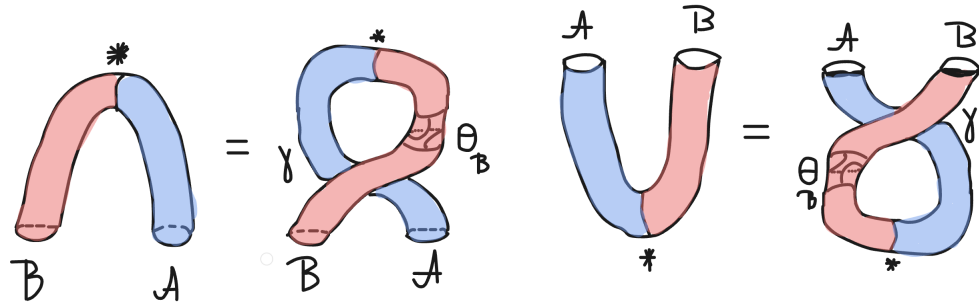
Definition 5.1. A *biexact* pairing $\mathcal{A} \dashv \mathcal{B}$ is a lax pairing whose 2-cells $\eta_{[2]}$ and $\varepsilon_{[2]}$ are reversible. A *biexact ribbon pairing* is a biexact pairing with a ribbon structure.

Definition 5.2. An *amphimonoid* $(\mathcal{A}, \mathcal{B})$ in a balanced monoidal bicategory $\mathcal{W}$ is defined as a pseudomonoid $(\mathcal{A}, \bigotimes, \text{true})$ and a pseudocomonoid $(\mathcal{B}, \bigotimes, \text{false})$ equipped with biexact pairing $\mathcal{A} \dashv \mathcal{B}$ (noted as $*$ below) and a pair of 2-cells:

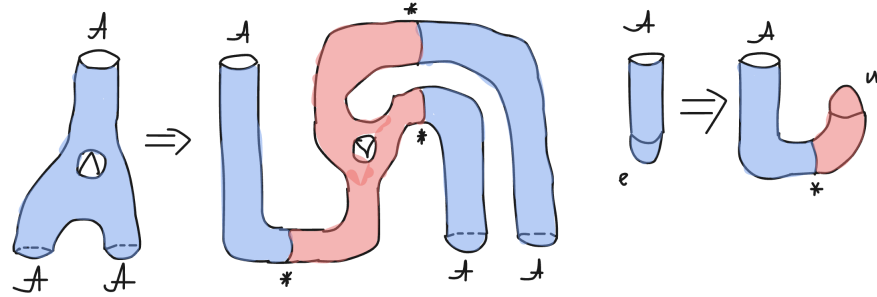


establishing a pseudomonoid equivalence between $(\mathcal{A}, \triangleleft, true)$ and a pseudomonoid on $\mathcal{B}$ induced by the structure of a biexact pairing.

Moreover, every amphimonoid induces a biexact ribbon pairing $\mathcal{B} \dashv \mathcal{A}$ defined as follows:



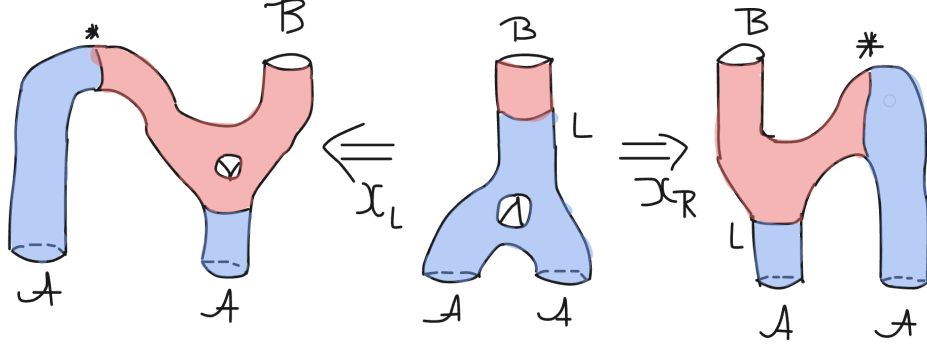
along with a pair of invertible 2-cells



Definition 5.3. A Frobenius amphimonoid $(\mathcal{A}, \mathcal{B}, L, R)$ consists of an amphimonoid $(\mathcal{A}, \mathcal{B})$ along with an adjunction

$$\mathcal{A} \begin{array}{c} \xrightarrow{L} \\ \perp \\ \xleftarrow{R} \end{array} \mathcal{B}$$

along with two invertible 2-cells



The 1-cell $L : \mathcal{A} \rightarrow \mathcal{B}$ is understood as a bracket $\langle a|b \rangle$ between $\mathcal{A}$ and $\mathcal{B}$ of the bicategory $\mathcal{V}$. Each side of the above equation may be seen as implementing curriffication set:

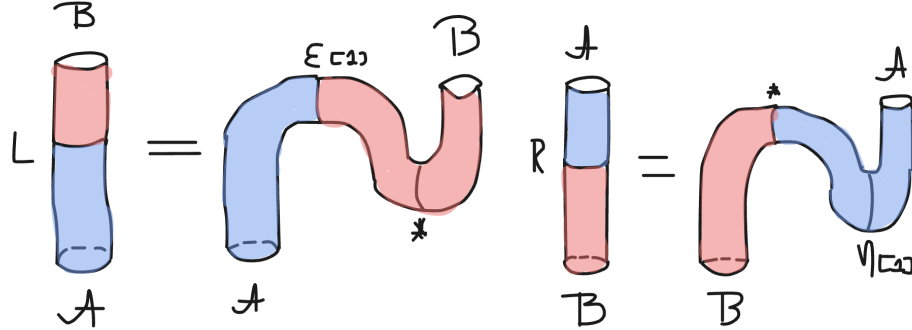
$$\begin{aligned}\chi_L : \langle a_1 \otimes a_2 | b \rangle &\Rightarrow \langle a_2 | a_1^* \vee b \rangle \\ \chi_R : \langle a_1 \otimes a_2 | b \rangle &\Rightarrow \langle a_1 | b \vee a_2^* \rangle\end{aligned}$$

Every helical dialogue category defines a Frobenius amphimonoid in **BiMon**, by taking $La := a \multimap \perp$ and $Ra := \perp \multimap a$.

Proposition 5.1. Let $(\mathcal{A}, \mathcal{B})$ be an amphimonoid in a balanced monoidal category $\mathcal{W}$, then there is a back-and-forth translation between the following two data:

- the helical Frobenius structures on the pseudomonoids $(\mathcal{A}, \otimes, true)$,
- the Frobenius structures (L, R, χ_L, χ_R) on the amphimonoids $(\mathcal{A}, \mathcal{B})$.

Proof. Given an amphimonoid $(\mathcal{A}, \mathcal{B})$, whose $\mathcal{A}$ -side is a Frobenius pseudomonoid $(\mathcal{A}, \otimes, true)$ with Frobenius form noted $||\cdot, \cdot||$, one defines the 1-cells L and R of the Frobenius amphimonoid as follows:



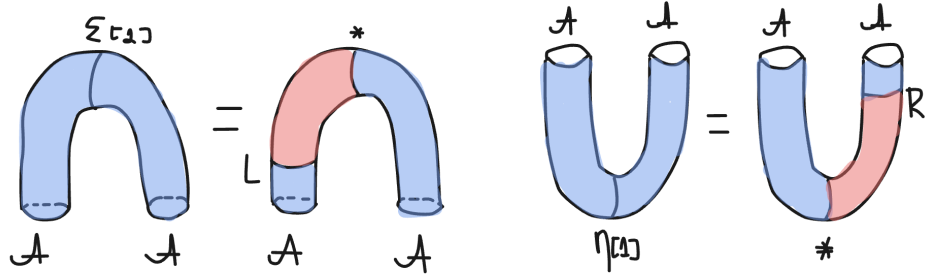
The definition of a lax pairing ensures $L \dashv R$ is indeed an adjunction in $\mathcal{W}$. The 2-cell χ_R is defined as:

$$||a_1 \otimes m, a_2|| \xRightarrow{\chi} ||a_1, m \otimes a_2||$$

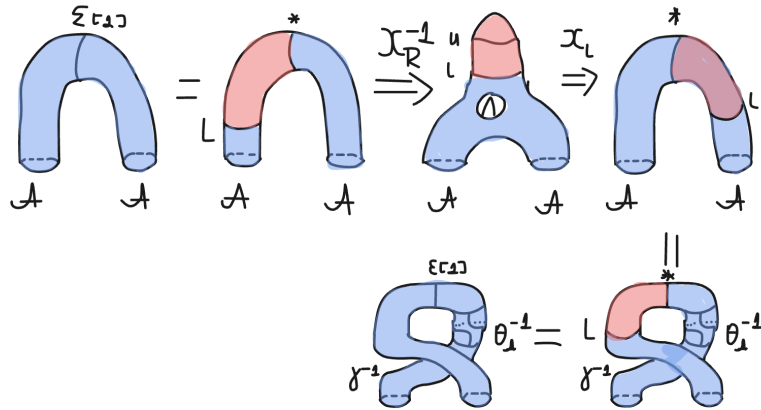
whilst χ_L is given as

$$||m \otimes a_1, a_2|| \xRightarrow{\text{helix}} ||a_2, m \otimes a_1|| \xRightarrow{\chi^{-1}} ||a_2 \otimes m, a_1|| \xRightarrow{\text{helix}^{-1}} ||a_1, a_2 \otimes m||$$

Conversely, let $(\mathcal{A}, \mathcal{B}, L, R)$ be a Frobenius amphimonoid in a balanced monoidal bicategory $\mathcal{W}$, a Frobenius form is defined the following way on the pseudomonoid $\mathcal{A}$:



The associativity law χ of the Frobenius form is defined using χ_R along with the coercion from the definition of a Frobenius amphimonoid and the 2-dimensional structure of the biexact pairing $\mathcal{A} \dashv \mathcal{B}$. One obtains a Frobenius pseudomonoid $(\mathcal{A}, \otimes, true)$, whose helical structure is defined using the currfication combinators χ_L and χ_R :

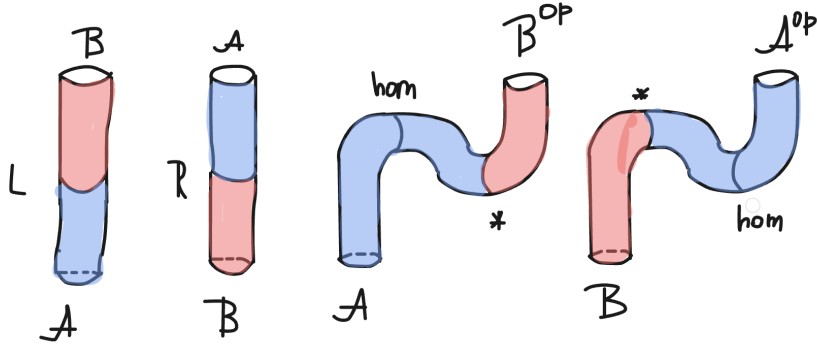


□

Now let us apply the above proposition to the specific monoidal bicategory **BiMod**, where every category $\mathcal{A}$ comes with the exact pairing $\mathcal{A} \dashv \mathcal{A}^{op}$ whose unit and counit are defined as the bimodule:

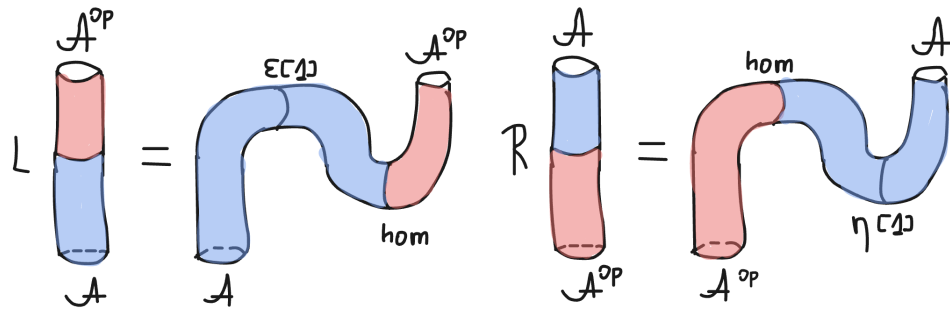
$$\begin{aligned} \text{hom} &: \mathcal{A}^{op} \times \mathcal{A} \rightarrow \text{Set} \\ \text{hom} &: (a_1, a_2) \mapsto \mathcal{A}(a_1, a_2) \end{aligned}$$

Theorem 5.1. (*First correspondence theorem*) A helical chirality is the same thing as a Frobenius amphimonoid in the bicategory **BiMod** whose 1-cells



are representable, i.e., images of functors along the 2-dimensional functor $(.)_\bullet : \text{Cat} \rightarrow \text{BiMod}$.

Theorem 5.2. (*Second correspondence theorem*) A helical dialogue category is the same thing as a helical Frobenius amphimonoid in the bicategory **BiMod** whose 1-cells



are representable, i.e., images of functors along the 2-dimensional functor $(.)_\bullet : \text{Cat} \rightarrow \text{BiMod}$.

Definition 5.4. A $*$ -autonomous pseudomonoid is a Frobenius pseudomonoid whose Frobenius form is based on the exact pairing $\mathcal{A} \dashv \mathcal{A}$.

Proposition 5.2. A pseudomonoid $(\mathcal{A}, m, e)$ is Frobenius if and only if it is equipped with 1-cell $l : \mathcal{A} \rightarrow I$ such that

$$\mathcal{A} \otimes \mathcal{A} \xrightarrow{m} \mathcal{A} \xrightarrow{l} I$$

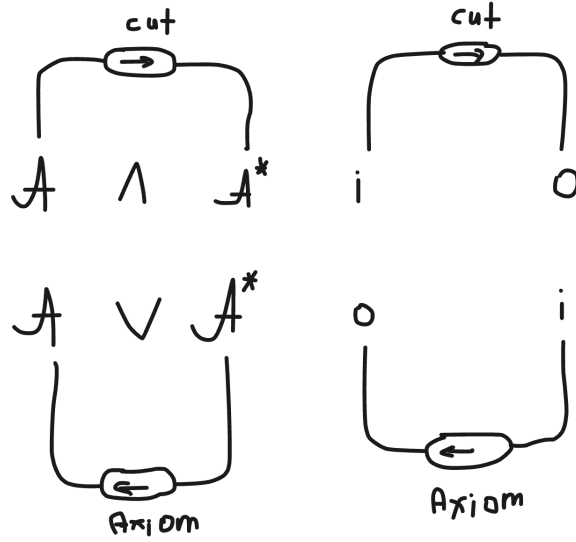
forms the unit $\eta_{[1]}$ in a lax pairing $\mathcal{A} \dashv \mathcal{A}$.

6 Ribbon Tensorial Logic: basic remarks

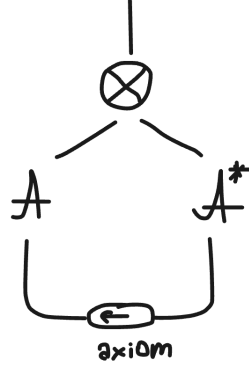
Category-theoretically, sequents $A \wedge A^* \vdash \mathbf{false}$ and $\mathbf{true} \vdash A \vee A^*$ define the following combinators:

$$\begin{aligned} \text{cut} &: A \wedge A^* \rightarrow \mathbf{false} \\ \text{axiom} &: \mathbf{true} \rightarrow A \vee A^* \end{aligned}$$

In the geometry of interaction, those combinators implement input-output channels:



In linear logic, one distinguishes the notions of *proof structure* and *proof net*. In multiplicative linear logic, a proof structure is a proof-like structure using the connectives $\otimes$ and $\wp$ along with the axiom and cut links. A proof tree is a proof structure generated by a derivation tree π . A small example:



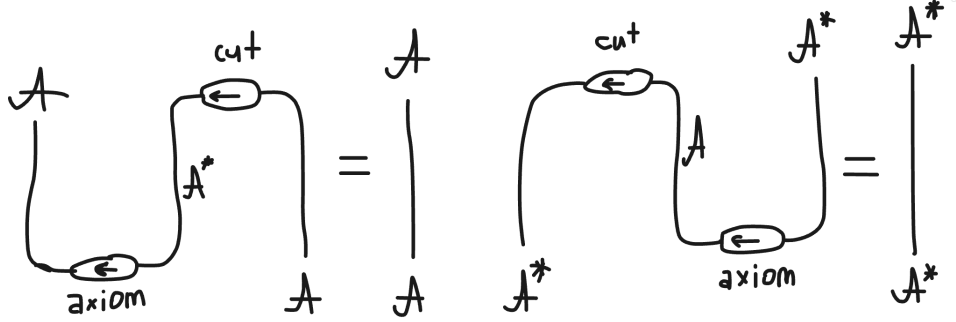
One defines proof nets as specific proof structures generated by generation trees in multiplicative linear logic. One can also replace the original definition of a proof net with the following one: a multiplicative proof net is a morphism of the free symmetric $*$ -autonomous category $*$ -**autonomous**($\mathcal{X}$) generated by a small category $\mathcal{X}$, typically, discrete. One can reformulate the notion of a proof structure to establish the connections between proof nets and proof structures. Recall that a compact closed category is a symmetric monoidal category with unit I where each object A is equipped with an object A^* and a pair of morphisms

$$\begin{aligned} \text{cut} &: A^* \otimes A \rightarrow I \\ \text{axiom} &: I \rightarrow A^* \otimes A \end{aligned}$$

such that the following conditions are satisfied:

$$\begin{array}{ccc} & A \otimes A^* \otimes A & \\ \text{axiom} \otimes A \nearrow & & \searrow A \otimes \text{cut} \\ A & \xlongequal{\quad} & A \end{array} \qquad \begin{array}{ccc} A^* & \xlongequal{\quad} & A^* \\ A^* \otimes \text{axiom} \searrow & & \nearrow \text{cut} \otimes A^* \\ & A^* \otimes A \otimes A^* & \end{array}$$

Consider the free compact-closed category **compact-closed**($\mathcal{X}$) for a small category $\mathcal{X}$ for defining proof structures. Those two triangles are visualised the following way in string diagrams:



This makes A^* right dual to A , written as $A \dashv A^*$. The operation $A \mapsto A^*$ defines an equivalence of categories $(.)^* : \mathcal{C}^{op} \rightarrow \mathcal{C}$. In other words, a compact-closed category is a $*$ -autonomous category where $\otimes$ and its opposite $\wp$ coincide up to symmetric monoidal equivalence.

The construction of the free compact closed category generated by $\mathcal{X}$ comes with a functor $\mathcal{X} \rightarrow \mathbf{compact-closed}(\mathcal{X})$. This functor can be lifted to the structure preserving functor between $*$ -autonomous categories:

$$[.] : *\text{-autonomous}(\mathcal{X}) \rightarrow \mathbf{compact-closed}(\mathcal{X})$$

such that the following triangle commutes:

$$\begin{array}{ccc}
 *\text{-autonomous}(\mathcal{X}) & \xrightarrow{[.]} & \mathbf{compact-closed}(\mathcal{X}) \\
 & \nwarrow \text{atoms} \quad \nearrow \text{atoms} & \\
 & \mathcal{X} &
 \end{array}$$

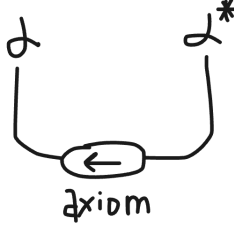
Thus, we have a categorical definition of a proof structure:

Definition 6.1. Let A and B be formulas of multiplicative linear logic with objects from $\mathcal{X}$ as atoms, a *proof structure* from A to B is a morphism $\Theta : [A] \rightarrow [B]$ in the free compact-closed category generated by $\mathcal{X}$. A proof structure for a formula A is a proof structure $\Theta : I \rightarrow [A]$.

Let us consider an example for the discrete category with a single element α . The functor $[.]$ applied to a formula A replaces each occurrence of $\otimes$ and $\wp$ with a tensor product. Therefore, $A \otimes A^*$ and $A \wp A^*$ are interpreted the same way:

$$[A \otimes A^*] = \alpha \otimes \alpha^* = [A \wp A^*]$$

By the definition, the morphism $\text{axiom} : I \rightarrow \alpha \otimes \alpha^*$ of the category $\mathbf{compact-closed}(\mathcal{X})$ is depicted as:



defines a proof structure θ of the formula $\alpha \otimes \alpha^*$ as well as a proof structure θ' of the formula $\alpha \wp \alpha^*$. The difference is that θ' is the image of a proof net $\pi : 1 \rightarrow \alpha \otimes \alpha^*$

$$\theta' = [\pi] : I \rightarrow [\alpha \otimes \alpha^*],$$

whereas θ is not.

There is a limitation that proof structures generally do not characterise proof nets 1 by 1. The analogous situation is that the identity and symmetry morphisms

$$\mathbf{id}, \mathbf{symm} : \perp \otimes \perp \rightarrow \perp \otimes \perp$$

generally do not have to coincide in *-autonomous categories. Consider the following derivation trees:

$$\frac{\frac{\frac{}{\vdash 1, \perp} \mathbf{ax}}{\vdash 1, 1, \perp \otimes \perp} \otimes}{\vdash 1 \wp 1, \perp \otimes \perp} \wp$$

$$\frac{\frac{\frac{}{\vdash 1, \perp} \mathbf{ax}}{\vdash 1, 1, \perp \otimes \perp} \otimes}{\vdash 1, 1, \perp \otimes \perp} \mathbf{exch} \quad \frac{}{\vdash 1 \wp 1, \perp \otimes \perp} \wp$$

Those tree are different only in applying the exchange rule in the second one, but they also define different morphisms in the free *-autonomous category, and therefore different proof nets π_1 and π_2 . But their images $[\pi_1]$ and $[\pi_2]$ in the free compact closed category coincide since $I = [1] = [\perp]$, and the following morphisms

$$\mathbf{id}, \mathbf{symm} : I \otimes I \rightarrow I \otimes I$$

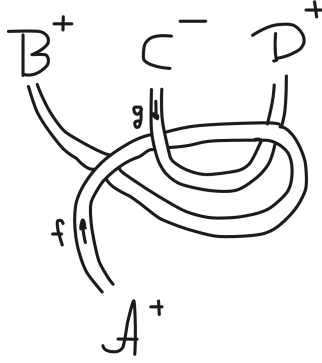
coincide in every symmetric monoidal category. Therefore, some information from the proofs nets π_1 and π_2 was lost after the translation. In more generic terms, this restriction can be formulated completely categorically is that the canonical functor $[\cdot] : \mathbf{*}\text{-autonomous}(\mathcal{K}) \rightarrow \mathbf{compact-closed}(\mathcal{K})$ is not faithful.

One can improve this restriction by taking a more general formalism than linear logic, tensorial logic, where negation is not involutive. To proceed further, one needs to take some ideas from knot theory and representation theory. The aim is to materialise links $[\pi]$ of a proof π .

The concept of a ribbon category is equipped with a coherence theorem, stating that the free ribbon category $\mathbf{ribbon}(\mathcal{X})$ has

- Objects: sequences $(A_1^{\epsilon_1}, \dots, A_n^{\epsilon_n})$ of signed objects of $\mathcal{X}$ where each ϵ_i is either $+$ or $-$.
- Morphisms: oriented ribbon tangles modulo topological deformation, where every open strand is coloured by a morphism of $\mathcal{X}$, and every closed strand is coloured by an equivalence class of morphisms modulo $f \circ g \sim g \circ f$ for a pair of morphisms $f : A \rightarrow B$ and $g : B \rightarrow A$.

For example, a morphism from (A^+) to (B^+, C^-, D^+) in $\mathbf{ribbon}(\mathcal{X})$ looks like this:

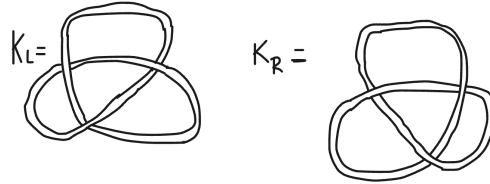


for $f : A \rightarrow B$ and $g : C \rightarrow D$ in $\mathcal{X}$.

Now consider the full and faithful functor $\mathcal{X} \rightarrow \mathbf{ribbon}(\mathcal{X})$ transporting every object A to the one-element sequence (A^+) . Every functor from the category $\mathcal{X}$ to a ribbon category $\mathcal{D}$ lifts as a structure-preserving $\langle \cdot \rangle$ (this is the universal property of the free algebras):

$$\begin{array}{ccc}
 \mathbf{ribbon}(\mathcal{X}) & \xrightarrow{\langle \cdot \rangle} & \mathcal{D} \\
 \swarrow \text{links} & & \nearrow \text{interpretation of links} \\
 & \mathcal{X} &
 \end{array}$$

Every topological ribbon knot defines a morphism $P : I \rightarrow I$, so its image $\langle P \rangle$ defines an invariant of a knot P modulo topological deformation. For instance, the functorial method allows establishing that the Jones polynomial $\langle P \rangle$ associated to a ribbon invariant. For example, the left trefoil K_L and the right trefoil K_R



are different modulo knot deformation as their Jones polynomials are different:

$$\begin{aligned}\langle K_L \rangle &= \frac{2}{x^2} + \frac{1}{x^4} + \frac{y^2}{x^2} \\ \langle K_R \rangle &= 2x^2 - x^4 + x^2y^2\end{aligned}$$

The braided concept of a dialogue category allows connecting proof theory and knot theory, so we will be working with balanced dialogue categories and ribbon tensorial logic. One has a similar coherence theorem from balanced dialogue categories and ribbon tensorial logic claiming that the free balance dialogue category generated by a category $\mathcal{X}$ has:

- objects: formulas of ribbon tensorial logic with atoms provided by the objects of $\mathcal{X}$,
- morphisms: the derivation trees π of proofs $A \vdash B$ factorised module the corresponding equational theory.

Now let us have a glance at the proof-as-tangle theorem closer. The next is to equip balanced dialogue categories with a topological content.

A pointed category $(\mathcal{C}, \perp)$ is a category with a distinguished object $\perp \in \mathcal{C}$. Alternatively, one can think of a pointed category as an S -algebra for the monad $S : \mathbf{Cat} \rightarrow \mathbf{Cat}$ sending each category $\mathcal{C}$ to $\mathcal{C} \sqcup \mathbf{1}$, where $\mathbf{1}$ is the terminal category. The unique object $\perp$ of $\mathbf{1}$ provides the single-out object of the pointed category $(\mathcal{C} \sqcup \mathbf{1}, \perp)$. Every category $\mathcal{C}$ induces a free ribbon category $\mathbf{ribbon}(\mathcal{C} \sqcup \mathbf{1})$ generated by $\mathcal{C} \sqcup \mathbf{1}$. By the construction, $\mathbf{ribbon}(\mathcal{C} \sqcup \mathbf{1})$ is monoidal and balanced. $\mathbf{ribbon}(\mathcal{C} \sqcup \mathbf{1})$ is also dialogue as the negations are given by:

$$\begin{aligned}x \multimap \perp &:= x^* \otimes \perp \\ \perp \multimap x &:= \perp \otimes x^*\end{aligned}$$

Note that, the following canonical morphism defining the strength of the double negation monad

$$(\perp \multimap (x \multimap \perp)) \otimes y \rightarrow \perp \multimap ((x \otimes y) \multimap \perp)$$

is an isomorphism in the resulting balanced dialogue category.

The unit of the monad S instantiated for $\mathcal{C}$

$$\mathbf{return}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C} \sqcup \mathbf{1}$$

induces the following functor:

$$\mathcal{C} \rightarrow \mathcal{C} \sqcup \mathbf{1} \rightarrow \mathbf{ribbon}(\mathcal{C} \sqcup \mathbf{1})$$

Therefore, there is a structure-preserving functor of balanced dialogue categories:

$$[\cdot] : \mathbf{balanced-dialogue}(\mathcal{C}) \rightarrow \mathbf{ribbon}(\mathcal{C} \sqcup \mathbf{1})$$

such that the following square commutes:

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{\text{return}_{\mathcal{C}}} & \mathcal{C} \sqcup \mathbf{1} \\
 \downarrow & & \downarrow \\
 \text{balanced-dialogue}(\mathcal{C}) & \xrightarrow{[\cdot]} & \text{ribbon}(\mathcal{C} \sqcup \mathbf{1})
 \end{array}$$

The functor $[\cdot]$ transports:

- formulas of ribbon tensorial logic into signed sequences of $\perp$'s and logical atoms provided by the objects of $\mathcal{C}$,
- proofs modulo the equational theory into ribbon tangles modulo topological deformation.

Definition 6.2. A tensorial proof net π of ribbon tensorial logic is a morphism in a free balanced dialogue category.

Definition 6.3. A tensorial proof structure of ribbon tensorial logic from a formula A to a formula B is a morphism $\Theta : [A] \rightarrow [B]$ in a free ribbon category.

Theorem 6.1. The canonical functor

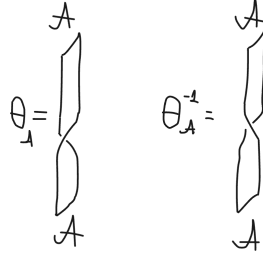
$$[\cdot] : \text{symmetric-dialogue}(\mathcal{C}) \rightarrow \text{compact-closed}(\mathcal{C} \sqcup \mathbf{1})$$

transporting tensorial proof nets to tensorial proof structures in commutative tensorial logic, is faithful.

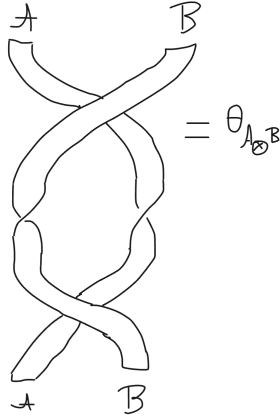
The theorem effectively means that in ribbon tensorial logic (as well as in commutative linear logic), derivation trees π are entirely characterised by their proof structures $[\pi]$. This theorem also connects proof theory with knot theory by providing a topological coherence theorem.

7 On Balanced Dialogue Categories and Ribbon Tensorial Logic

Recall that in balanced categories, the twist θ_A is visualised as the ribbon A twisted positively in the trigonometric direction with an angle 2π , whereas its inverse θ_A^{-1} is depicted the same way as A , but twisted negatively with an angle -2π :



So one can reword the commutative square from the definition of a balanced category is visualised as follows:



Definition 7.1. A *balanced dialogue category* is a dialogue category equipped with twists and braiding.

One can extract an example of a balanced monoidal category from algebra, in particular, from the representation of quantum groups. The category $\mathbf{Mod}(H)$ of (finite or infinite dimensional) H -modules associated to a ribbon Hopf algebra H . The full subcategory of rigid objects (i.e., those ones that have a right dual) in $\mathbf{Mod}(H)$ (or in any other balanced dialogue category) is a ribbon category. Typically, the category $\mathbf{Mod}_f(H)$ of finite dimensional H -modules associated to a ribbon Hopf algebra H defines a ribbon category.

Recall that the braid group $\mathbf{B}_n$ on $n < \omega$ strands is presented by generators $\{\sigma_i \mid 1 \leq i \leq n-1\}$ and the following equations:

$$\begin{aligned} \sigma_i \circ \sigma_{i+1} \circ \sigma_i &= \sigma_{i+1} \circ \sigma_i \circ \sigma_{i+1} \\ \sigma_i \circ \sigma_j &= \sigma_j \circ \sigma_i \text{ for either } |j-i| \leq 2 \end{aligned}$$

There is a left action:

$$\triangleleft: \mathbf{B}_n \times [n] \rightarrow [n]$$

of the group $\mathbf{B}_n$ to the set $[n] = \{1, \dots, n\}$ of strands induced by the group homomorphisms $\mathbf{B}_n \rightarrow \mathbf{S}_n$, where $\mathbf{S}_n$ is the symmetry group on n elements.

This action defines a wreath product of $\mathbf{B}_n$ on the additive group $(\mathbb{Z}, +, 0)$. The resulting group is $\mathbf{G}_n$, the ribbon group on n strands. It is defined on the generators $\{\sigma_i \mid 1 \leq i \leq n-1\}$ and $\{\theta_i \mid 1 \leq i \leq n\}$ and the related from the definition of a braid group along with the following ones:

- $\sigma_i \circ \theta_i = \theta_{i+1} \circ \sigma_i$,
- $\sigma_i \circ \theta_{i+1} = \theta_i \circ \sigma_i$,
- $\sigma_i \circ \theta_j = \theta_j \circ \sigma_i$ for either $j < i$ or $j \geq i+2$.

One can conceptually rephrase ribbon groups as follows. The groupoid $\mathcal{G}$ defined as the disjoint sum of groupoids $S\mathbf{G}_n$ coincides with the free balanced category generated by the terminal category $\mathbf{1}$. The group $\mathbf{G}_n$ can be alternatively defined as $\mathcal{G}(n, n)$ where

$$n = 1 \otimes \dots \otimes 1$$

is the n -fold tensor product of the generator 1 in the terminal category $\mathbf{1}$. Therefore, we have a family of group homomorphisms

$$\otimes : \mathbf{G}_p \times \mathbf{G}_q \rightarrow \mathbf{G}_{p+q}$$

reflecting the monoidal structure of the balanced category $\mathcal{G}$. Moreover, the above defined action

$$\triangleleft : \mathbf{G}_n \times [n] \rightarrow [n]$$

where each θ_i acts trivially $[n] = \{1, \dots, n\}$ in the sense that $\theta_i \triangleleft k = k$ for each $k \in [n]$.

8 The Sequent Calculus and its commuting conversions

The formulas of ribbon tensorial logic is generated by the following grammar:

$$A, B ::= \mathbf{1} \mid \alpha \mid \perp \mid (A \otimes B) \mid (A \multimap \perp) \mid (\perp \multimap A)$$

where α ranges over objects from a small category $\mathcal{X}$.

The sequents have the form of:

$$A_1, \dots, A_m \vdash B$$

where $A_1, \dots, A_m$ is a sequence of formula and B is a single formula. The inference rules are given in Figure 1.

Figure 1: Ribbon Tensorial Logic

$$\begin{array}{c}
\frac{f : A \rightarrow B \in \mathcal{X}}{A \vdash B} \mathbf{ax} \qquad \frac{\Gamma \vdash A \quad \Delta, A, \Pi \vdash B}{\Delta, \Gamma, \Pi \vdash B} \mathbf{cut} \\
\\
\frac{\Gamma_1, \Gamma_2 \vdash A}{\Gamma_1, \mathbf{1}, \Gamma_2 \vdash A} \mathbf{1}_L \qquad \frac{}{\vdash \mathbf{1}} \mathbf{1}_R \\
\\
\frac{\Gamma_1, A, B, \Gamma_2 \vdash A}{\Gamma_1, A \otimes B, \Gamma_2 \vdash A} \otimes_L \qquad \frac{\Gamma \vdash A \quad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B} \otimes_R \\
\\
\frac{A, \Gamma \vdash \perp}{\Gamma \vdash A \multimap \perp} (\multimap \perp)_L \qquad \frac{\Gamma \vdash A}{\Gamma, A \multimap \perp \vdash \perp} (\multimap \perp)_R \\
\\
\frac{\Gamma, A \vdash \perp}{\Gamma \vdash \perp \multimap A} (\perp \multimap)_L \qquad \frac{\Gamma \vdash A}{\perp \multimap A, \Gamma \vdash \perp} (\perp \multimap)_R \\
\\
\frac{A_1, \dots, A_n \vdash B \quad g \in \mathbf{G}_n}{A_{g \triangleleft 1}, \dots, A_{g \triangleleft n} \vdash B}
\end{array}$$

Ribbon tensorial logic is inspired by knot theory, so one can bear this in mind in designing the equational theory on derivation trees based on the commuting conversions. The actual difference is that the exchange rule is labelled by the elements of the ribbon group, so one needs to pay extra attention in treating the commuting conversions involving the ribbon exchange rule. The challenge in such commuting conversions is to label the exchange rules appropriately on each side π_1 and π_2 in $\pi_1 \rightsquigarrow \pi_2$. As an example, consider the following derivation tree

$$\frac{\frac{\pi_1}{\dots} \quad \frac{\pi_2}{\dots}}{\frac{A_1, \dots, A_n \vdash A \quad \frac{\Gamma_1, B, A, \Gamma_2 \vdash C}{\Gamma_1, A, B, \Gamma_2 \vdash C} p \otimes \sigma \otimes q}{\Gamma_1, A_1, \dots, A_n, B, \Gamma_2 \vdash C} \mathbf{cut}}$$

that we convert into the following derivation tree:

$$\frac{\frac{\pi_1}{\dots} \quad \frac{\pi_2}{\dots}}{\frac{A_1, \dots, A_n \vdash A \quad \frac{\Gamma_1, B, A, \Gamma_2 \vdash C}{\Gamma_1, B, A_1, \dots, A_n \vdash C} p \otimes \sigma_{n,1} \otimes q}{\Gamma_1, A_1, \dots, A_n, B \vdash C} \mathbf{cut}}$$

where p and q are the lengths of Γ_1 and Γ_2 respectively and $\sigma_{m,n}$ is the positive braid permutting m strands above n strands.

Another example is the conversion that transforms the below derivation tree

$$\frac{\frac{\pi_1}{\dots} \quad \frac{\frac{\pi_2}{\dots} \quad \frac{\Gamma_1, A, \Gamma_2 \vdash C}{\Gamma_1, A, \Gamma_2 \vdash C}}{A_1, \dots, A_n \vdash A} \quad \frac{\Gamma_1, A, \Gamma_2 \vdash C}{\Gamma_1, A, \Gamma_2 \vdash C} p \otimes \theta \otimes q}{\Gamma_1, A_1, \dots, A_n, \Gamma_2 \vdash C} \text{cut}$$

into the following one:

$$\frac{\frac{\pi_1}{\dots} \quad \frac{\pi_2}{\dots}}{A_1, \dots, A_n \vdash A} \quad \frac{\Gamma_1, A, \Gamma_2 \vdash C}{\Gamma_1, A, \Gamma_2 \vdash C} \text{cut}}{\frac{\Gamma_1, A_1, \dots, A_n, \Gamma_2 \vdash C}{\Gamma_1, A_1, \dots, A_n, \Gamma_2 \vdash C} p \otimes \theta \langle n \rangle \otimes q}$$

where $\langle n \rangle$ is the positive twist on n strands. Categorically, these two commuting conversions claim that the braiding σ and the twist θ are natural respectively.

Yet another example is the following conversion for any $g, h \in \mathbf{G}_n$ that transforms the following derivation:

$$\frac{\frac{\pi}{\dots} \quad \frac{A_1, \dots, A_n \vdash A}{A_{h \triangleleft 1}, \dots, A_{h \triangleleft n} \vdash A} [h]}{A_{g \triangleleft (h \triangleleft 1)}, \dots, A_{g \triangleleft (h \triangleleft n)} \vdash A} [g]$$

into the following one:

$$\frac{\frac{\pi}{\dots} \quad \frac{A_1, \dots, A_n \vdash A}{A_{g \circ h \triangleleft 1}, \dots, A_{g \circ h \triangleleft n} \vdash A} [g \circ h]}{A_{g \circ h \triangleleft 1}, \dots, A_{g \circ h \triangleleft n} \vdash A} [g \circ h]$$

And there is a similar conversion for the identity, where the following derivation

$$\frac{\frac{\pi}{\dots} \quad \frac{A_1, \dots, A_n \vdash A}{A_{e \triangleleft 1}, \dots, A_{e \triangleleft n} \vdash A} [e]}{A_{e \triangleleft 1}, \dots, A_{e \triangleleft n} \vdash A} [e]$$

into the following one

$$\frac{\pi}{\dots} \quad \frac{A_1, \dots, A_n \vdash A}{A_1, \dots, A_n \vdash A}$$

Another example is induced by the coherence diagram for braiding, where the following derivation

$$\begin{array}{c}
\pi \\
\vdots \\
\hline
\Gamma, A, B, C, \Pi \vdash D \\
\hline
\Gamma, A \otimes B, C, \Pi \vdash D \quad \otimes_L \\
\hline
\Gamma, C, A \otimes B, \Pi \vdash D \quad p \otimes \sigma \otimes q
\end{array}$$

into the following one

$$\begin{array}{c}
\pi \\
\vdots \\
\hline
\Gamma, A, B, C, \Pi \vdash D \\
\hline
\Gamma, A, C, B, \Pi \vdash D \quad p \otimes 1 \otimes \sigma \otimes q \\
\hline
\Gamma, C, A, B, \Pi \vdash D \quad p \otimes \sigma \otimes 1 \otimes q \\
\hline
\Gamma, C, A \otimes B, \Pi \vdash D \quad \otimes_L
\end{array}$$

Similarly, there is a conversion induced by the coherence conditions for twists, the following derivation

$$\begin{array}{c}
\pi \\
\vdots \\
\hline
\Gamma, A, B, \Pi \vdash D \\
\hline
\Gamma, A \otimes B, \Pi \vdash D \quad \otimes_L \\
\hline
\Gamma, A \otimes B, \Pi \vdash D \quad p \otimes \theta \otimes q
\end{array}$$

is identified with the following one:

$$\begin{array}{c}
\pi \\
\vdots \\
\hline
\Gamma, A, B, \Pi \vdash D \\
\hline
\Gamma, B, A, \Pi \vdash D \quad p \otimes \sigma \otimes q \\
\hline
\Gamma, B, A, \Pi \vdash D \quad p \otimes \theta \otimes \theta \otimes q \\
\hline
\Gamma, A, B, \Pi \vdash D \quad p \otimes \sigma \otimes q \\
\hline
\Gamma, A \otimes B, \Pi \vdash D \quad \otimes_L
\end{array}$$

Theorem 8.1. Every derivation tree in ribbon tensorial logic is equivalent to a cut-free derivation tree modulo commuting conversions.

9 Focusing Theorem

The commuting conversions also allow establishing the focusing theorem in order to ensure that every cut-free derivation π to its normal form π_{nf} by applying a finite number of commuting conversions.

We start with a number of derivation trees $\pi_1 : \Gamma_1 \vdash A_1, \dots, \pi_n : \Gamma_n \vdash A_n$ where all the formulas from $\Gamma_1, \dots, \Gamma_n$ are either atomic or negated. One then apply the following phases one after the other to get a derivation tree π of the same form, whose sequent $\Gamma \vdash A$ has all the formulas from Γ either atomic or negated:

1. A left introduction rule for the left/right negation produces a sequent whose conclusion formula is $\perp$,
2. A series of exchange rules permuting formulas from a context,
3. A series of the left introduction rules for $\mathbf{1}$ and $\otimes$ that produces a sequent where at most one formula is the context is either atomic or negated,
4. A right introduction of the left negation or the right negation or an axiom rule where all formulas are either negated or atomic,
5. A series of the right introduction rules for $\mathbf{1}$ and $\otimes$,
6. A series of exchange rules permutting either atomic or negated formulas.

Theorem 9.1. Every cut-free derivation π is equivalent to a focused derivation tree π_{nf} by applying a finite number of commuting conversions of ribbon tensorial logic.

Given a functor $\mathcal{X} \rightarrow \mathcal{D}$ from the category of atoms $\mathcal{X}$ to a balanced dialogue category. Then one has:

Theorem 9.2. Soundness

Every derivation π of a sequent:

$$A_1 \otimes \dots \otimes A_n \vdash A$$

in ribbon tensorial logic is interpreted as a morphism

$$[\pi] : [A_1] \otimes \dots \otimes [A_n] \vdash [A]$$

in the balanced dialogue category $\mathcal{D}$. Besides, the interpretation $[\pi]$ provides the invariant of a derivation tree.

The interpretation $[\pi]$ of a derivation tree π is defined by structural induction. In particular, the interpretation of the exchange rule

$$\frac{A_1, \dots, A_n \vdash B \quad g \in \mathbf{G}_n}{A_{g \triangleleft 1}, \dots, A_{g \triangleleft n} \vdash B}$$

is interpreted the following way:

$$\frac{[A_1] \otimes \dots \otimes [A_n] \xrightarrow{[\pi]} [B] \quad g \in \mathbf{G}_n}{A_{g \triangleleft 1} \otimes \dots \otimes A_{g \triangleleft n} \xrightarrow{\theta} [A_1] \otimes \dots \otimes [A_n] \xrightarrow{[\pi]} [B]}$$

where θ is a morphism associated to $g \in \mathbf{G}_n$ acting on $[A_1] \otimes \dots \otimes [A_n]$.

10 Proof-as-Tangle theorem

There is another way to formulate the soundness theorem. One can state that the free balanced dialogue category $\mathbf{dialogue}(\mathcal{X})$ generated by $\mathcal{X}$ coincides with a category of tensorial formulas and derivation trees modulo the equational theory of ribbon tensorial logic. Then one can deduce the functor

$$[\cdot] : \mathbf{dialogue}(\mathcal{X}) \rightarrow \mathbf{ribbon}(\mathcal{X} \sqcup \mathbf{1})$$

that transports every tensorial proof net π into a topological tangle $[\pi]$.

Theorem 10.1. The functor $[\cdot]$ is faithful.

Proof. Consider two cut-free derivation trees $\pi_1 : A \vdash B$ and $\pi_2 : A \vdash B$ in tensorial ribbon logic inducing the same tangle $[\pi_1] = [\pi_2]$ modulo topological deformation in the free ribbon category $\mathbf{ribbon}(\mathcal{X} \sqcup \mathbf{1})$. We show that $\pi_1 \rightsquigarrow \pi_2$ and conclude. We proceed by induction on the number of strands in the tangle. By the focusing theorem, we know that the proofs π_1 and π_2 are equal modulo the equational theory to the following derivations:

$$\frac{\begin{array}{c} \pi'_1 \\ \dots \end{array}}{A_1, \dots, A_n \vdash B} \quad \frac{\begin{array}{c} \pi'_2 \\ \dots \end{array}}{A_1, \dots, A_n \vdash B}$$

where each A_i is either a negation or an atom, followed by the same sequence of left introduction of tensor or unit. Suppose that $B = \perp$, then $\perp$ was either introduced by:

- the left introduction of a left negation $X \multimap \perp$,
- the left introduction of a right negation $\perp \multimap X$.

By symmetry, we may suppose that the last rule introduces a left negation $X \multimap \perp$. In this case, π'_1 is equal to

$$\frac{\frac{\begin{array}{c} \pi''_1 \\ \dots \end{array}}{X_1, \dots, X_{n-1} \vdash X}}{X_1, \dots, X_{n-1}, X \multimap \perp \vdash \perp} \multimap_L \quad [g]$$

In turn, π'_2 is equal to the following proof:

$$\begin{array}{c}
\pi_2'' \\
\vdots \\
\hline
\frac{Y_1, \dots, Y_{n-1} \vdash X}{Y_1, \dots, Y_{n-1}, X \multimap \perp \vdash \perp} \multimap_L \\
\hline
\frac{}{A_1, \dots, A_n \vdash \perp} [h]
\end{array}$$

So we conclude that

$$\begin{array}{c}
\pi_1'' \\
\vdots \\
\hline
\frac{X_1, \dots, X_{n-1} \vdash X}{X_1, \dots, X_{n-1}, X \multimap \perp \vdash \perp} \multimap_L
\end{array}$$

induced the same topological tangle as the derivation tree below:

$$\begin{array}{c}
\pi_2'' \\
\vdots \\
\hline
\frac{Y_1, \dots, Y_{n-1} \vdash X}{Y_1, \dots, Y_{n-1}, X \multimap \perp \vdash \perp} \multimap_L \\
\hline
\frac{A_1, \dots, A_n \vdash \perp}{X_1, \dots, X_{n-1}, X \multimap \perp \vdash \perp} [g^{-1}]
\end{array}$$

All the formulas A_i are either negated or atoms, and thus $g^{-1} \circ h \in \mathbf{G}_n$ is of the form $f \otimes 1$. From this follows that the proof

$$\begin{array}{c}
\pi_1'' \\
\vdots \\
\hline
X_1, \dots, X_{n-1} \vdash X
\end{array}$$

has

$$\begin{array}{c}
\pi_2'' \\
\vdots \\
\hline
\frac{X_1, \dots, X_{n-1} \vdash X}{Y_1, \dots, Y_{n-1} \vdash Y} [f]
\end{array}$$

They are therefore equal by the induction hypothesis. π_1' is thus equal to the proof

$$\begin{array}{c}
\pi_2'' \\
\vdots \\
\hline
\frac{Y_1, \dots, Y_{n-1} \vdash X}{Y_1, \dots, Y_{n-1}, X \multimap \perp \vdash \perp} \multimap_L \\
\hline
\frac{X_1, \dots, X_{n-1}, X \multimap \perp \vdash \perp}{A_1, \dots, A_n \vdash \perp} [f \otimes 1] \\
\hline
[g]
\end{array}$$

that can be rewritten into the proof:

$$\frac{\displaystyle \frac{\displaystyle \frac{\pi_2''}{\dots}}{Y_1, \dots, Y_{n-1} \vdash X}}{Y_1, \dots, Y_{n-1}, X \multimap \perp \vdash \perp} \multimap_L \frac{}{A_1, \dots, A_n \vdash \perp} [h]$$

and therefore π_1 and π_2 are equal. $\square$

References

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